A3-PS007-TC223

Privacy-Aware IPv6 IoT Security Monitoring Across Address Rotation

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Problem Description

- IPv6 Address Cycling: The regular cycling of addresses makes IoT device tracking tough.
- Unauthorized Devices: Devices that may not have the right access may join the IoT network.
- Device Identification: Cycling of IPv6 addresses may make it difficult to identify the devices.
- Privacy: Network monitoring must ensure that device identity and other information remain confidential.
- Anomalies: Any suspicious network traffic and activity must be flagged.
- Real Time Alerts: Security threats must be detected and reported in real time.
- Network Monitoring: It is necessary for the administrator to see all the IPv6 IoT devices connected.
- Risk Assessment: Detected actions must be converted into risks.
- Monitoring Portal: A portal-based web application must show all devices, IPv6 addresses, alerts, and security risks.

Background

The fast development of the Internet of Things (IoT) has led to billions of devices being interconnected through the use of networks. IPv6 becomes increasingly important for IoT due to the possibility of providing a huge amount of addresses for the numerous devices that should be able to connect to networks.

However, due to the IPv6 privacy features, devices may change their addresses from time to time. Even though address rotation helps increase privacy, it makes it more complicated for the IoT security monitoring systems to recognize the device after the change.

In addition, the IoT networks may include devices with different capabilities, software, and security settings. In case of an intrusion, an unauthorized device will communicate in a suspicious way, generate abnormal traffic, and try to interact with other devices.

That is why it is necessary to develop a security solution that would be capable of keeping track of the IoT devices despite the IPv6 address changes while decreasing the unnecessary disclosure of the device identity and sensitive data.

The aim of our project is the development of a web-based, privacy-aware IPv6 IoT security monitoring platform. The platform includes IPv6 monitoring, device identity correlation, behavioral analysis, rogue-device detection risks.

Literature & RFC Review

- IPv6 Stateless Address Autoconfiguration – RFC 4862: Defines SLAAC, through which IPv6 hosts can automatically configure addresses and perform Duplicate Address Detection. This establishes the basic mechanism behind dynamic IPv6 addressing.

- Temporary IPv6 Addresses – RFC 8981: Defines temporary IPv6 addresses with randomized interface identifiers to improve privacy and reduce long-term address-based tracking. However, frequent address changes create challenges for continuous IoT device monitoring and identity correlation.

- IPv6 Security Considerations – RFC 9099: Provides operational security guidance for IPv6 networks, including the need for appropriate ACLs, firewalls, host security, and protection against IPv6-specific threats such as rogue Router Advertisements and Neighbor Discovery attacks.

- Literature Gap: Existing standards provide mechanisms for IPv6 configuration, privacy, and network security, but they do not by themselves provide a unified, web-based IoT monitoring system that correlates devices across privacy-driven address changes.

- Our Research Direction: We build on these standards by combining IPv6 monitoring, privacy-aware device correlation, rogue-device detection, behavioral anomaly analysis, risk scoring, and real-time web-based security alerts.

Proposed Solution

PRIV6-SHIELD is a privacy-aware, web-based IPv6 IoT security monitoring system designed to detect network threats while avoiding permanent device tracking.

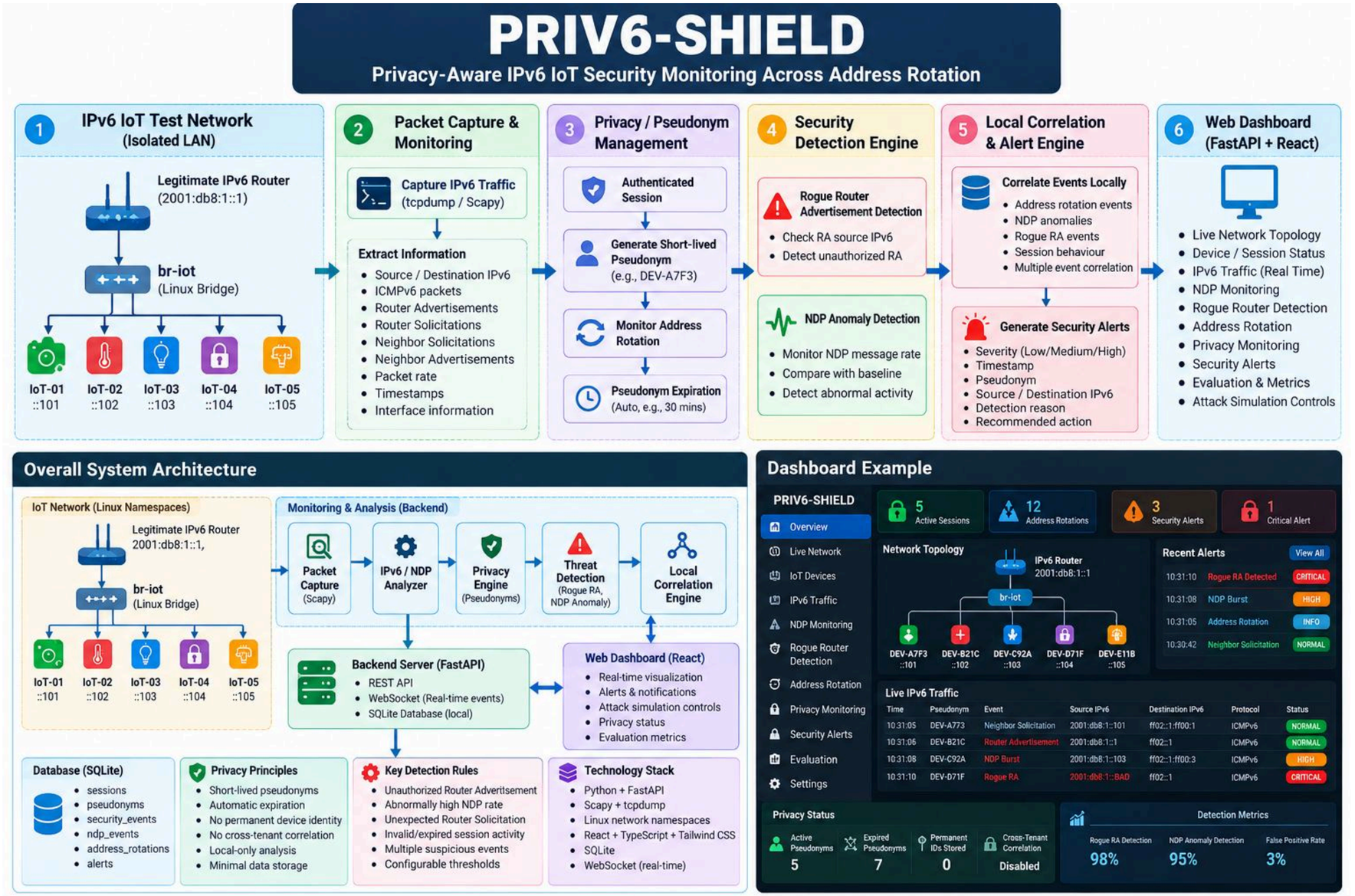
- Isolated IPv6 IoT Network: Create one local IPv6 LAN containing five emulated IoT devices and a legitimate IPv6 router. PS-007
- Privacy-Preserving Identity: Replace permanent device identifiers with short-lived local pseudonyms that automatically expire.
- IPv6 Address Rotation: Allow IoT devices to periodically change their IPv6 addresses while maintaining authenticated local sessions.
- Real-Time Monitoring: Capture and analyze IPv6 and ICMPv6 traffic locally using a monitoring pipeline.
- Rogue Router Detection: Identify unauthorized Router Advertisement messages and generate high-severity security alerts.
- NDP Anomaly Detection: Establish normal Neighbor Discovery traffic levels and detect abnormal message-rate increases. PS-007
- Local Event Correlation: Correlate address rotations, sessions, NDP activity, and security events locally without relying on permanent cross-device tracking.
- Web-Based Security Dashboard: Provide live network topology, device/session status, IPv6 activity, alerts, address rotations, and privacy status.
- Privacy Verification: Verify pseudonym expiration and ensure that cross-tenant correlation and permanent identity recovery are not performed. PS-007
- Performance Evaluation: Measure detection rate, false positives, attack detection, address-rotation handling, and privacy-related outcomes.

Optimization Proposed

PRIV6-SHIELD can be optimized to provide faster threat detection while reducing unnecessary data collection and false alerts.

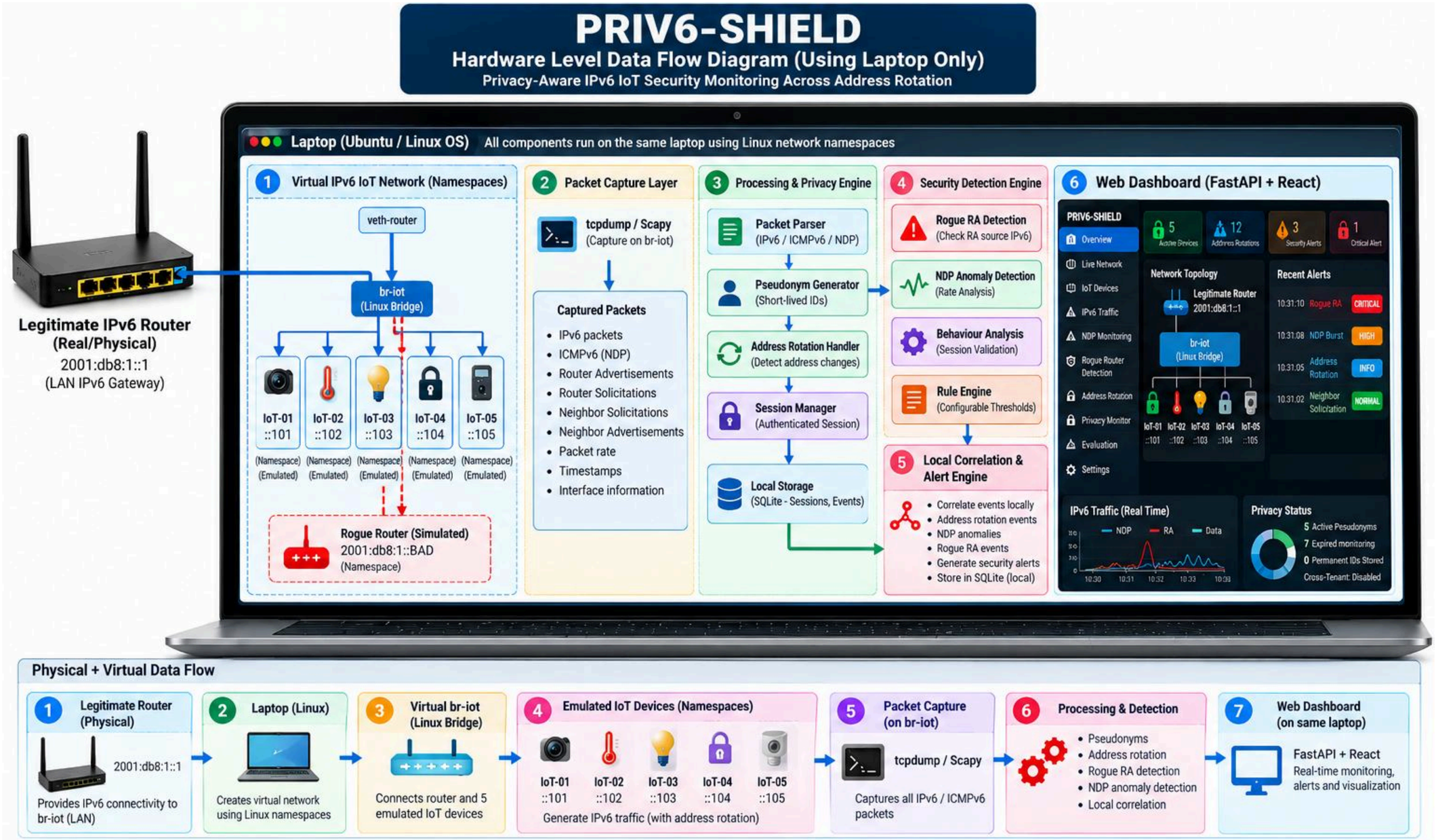
- Lightweight Local Processing: Analyze IPv6 and ICMPv6 traffic locally instead of sending raw network data to external services.
- Adaptive NDP Thresholds: Dynamically adjust the normal NDP traffic baseline according to the current IoT network activity to reduce false positives.
- Efficient Pseudonym Management: Use short-lived pseudonyms with automatic expiration so unnecessary identity mappings are not retained.
- Event-Based Monitoring: Process important security events rather than storing every packet indefinitely, reducing processing and storage requirements.
- Fast Rogue-RA Detection: Maintain a local list of authorized router information so unauthorized Router Advertisements can be detected quickly.
- Local Event Correlation: Combine related events such as address rotation and abnormal NDP activity before generating an alert, preventing duplicate notifications.
- Real-Time Dashboard Optimization: Use WebSockets to send only relevant security events to the dashboard instead of repeatedly refreshing the complete dataset.
- Privacy-by-Design Storage: Store only the minimum information required for security analysis and automatically remove expired session/pseudonym information.
- Scalable Architecture: Keep packet capture, privacy management, detection, backend, and dashboard as separate modules so additional IoT devices and detection rules can be added later.
- Performance Evaluation: Continuously evaluate detection rate, false-positive rate, pseudonym expiration, and address-rotation handling to tune the system.

Solution Architecture & Design



Data Flow Diagram

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Deployment Timeline

The PRIV6-SHIELD deployment will be completed in a 14-day phased development plan. During Days 1–2, the team will prepare the Linux environment, configure the legitimate IPv6 router, and create the isolated IPv6 IoT network. Days 2–3 will focus on creating and connecting five emulated IoT devices. During Days 3–4, IPv6, ICMPv6, and Neighbour Discovery traffic monitoring will be implemented using packet-capture tools. Days 4–6 will develop short-lived pseudonyms, authenticated sessions, and IPv6 address-rotation mechanisms. Days 6–8 will implement rogue Router Advertisement detection, abnormal NDP-rate detection, and local event correlation. Days 8–10 will integrate the FastAPI backend, local storage, real-time communication, and web-based security dashboard. Days 10–12 will test normal traffic, address rotation, rogue Router Advertisements, and NDP anomalies while optimizing detection and reducing false positives. Days 12–13 will evaluate detection performance and privacy requirements. Day 14 will complete system integration, documentation, presentation materials, and the final live demonstration.

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9. FastAPI Documentation, Python framework for building APIs and real-time backend services.
10. React Documentation, JavaScript library for developing interactive web-based user interfaces.

Thank You for Your Attention!